HRN-v2: A Hierarchical Resonant Network for Sequential MNIST under Strict Biological-Plausibility Constraints

(Analog and Strict-Spiking Variants)

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Abstract

We present **HRN-v2**, a hierarchical resonant neural network whose every mechanism — forward dynamics, credit assignment, plasticity rule, and readout — is local in time and local in space. The substrate is a population of complex damped-rotation oscillators (one ODE per neuron, no recurrent inter-neuron coupling within a stage); the credit signal is a forward-only eligibility trace; the readout is a class-pool tail-energy projection; supervision is DECOLLE-style per-stage local cross-entropy. There is no backpropagation through time, no surrogate-gradient spike emission, and no symmetric backward pathway. Stages are stacked with a sparse class-aligned fan-in and a strict frequency-band hierarchy (faster $\rightarrow$ slower across stages). We report two variants:

1. **Analog variant** (continuous tail-energy readout, no spike emission). On Sequential MNIST (SMNIST, 784-step single-channel pixel sequence, 10-way classification, 10k training subset) the 3-stage HRN-v2 reaches **89.9%** test accuracy.
2. **Strict-spiking variant** (every neuron emits binary spikes; inter-neuron synapses carry a single bit per timestep). The spike-emission rule is stochastic Bernoulli of $\sigma(\beta(|z|^2-\theta))$ and the gradient flows through the σ' that is the *natural* derivative of the Bernoulli expected rate (not an ad-hoc surrogate of a Heaviside). A 5-stage spiking HRN-v2 reaches **81.35%** on the same SMNIST 10k benchmark, and **60.6%** on Spiking Heidelberg Digits (SHD, 20-class spoken-digit, 4k train).

The strict-spiking version inherits all the bio-plausibility constraints of the analog version and additionally restricts inter-neuron messages to a single binary spike per timestep. The 8.5pp gap between the two SMNIST numbers is the empirical cost of that restriction. Several mechanisms intended to close this gap (subtractive reset, intra-stage recurrence with random init, top-down predictive feedback) were tested, with the diagnosis: the LIF-style amplitude-consume reset is incompatible with oscillator phase dynamics, and naive intra-stage recurrence destabilises deep high-α stages. Pure depth scaling (5 stages with strict frequency-band hierarchy) is the clean win.

1 Introduction and motivation

We adopt the working hypothesis that biological neural computation is naturally expressed in the language of *resonance*:

- *computation* is the act of an internal oscillator phase-locking with structure in its driving signal;
- *learning* is the local tuning of dynamical-system parameters (frequency, damping, input weights) of each oscillator;

- *memory* is the stable resonant configuration the system settles into for a class of inputs;
- *decision-making* is the act of one resonant mode dominating another at readout time.

A natural consequence of this view is *recursive decomposition*: a hard problem is broken into modes; each mode is handled by a pool of neurons; each pool itself decomposes its sub-problem into smaller modes at a slower time-scale. The architecture of HRN-v2 is built directly on this principle.

The substrate is a strictly faithful version of the linear damped complex oscillator from [1] (the per-neuron model used in `resonant_self_organizing_layer.py`); we deliberately remove the cubic Stuart–Landau term, the spike emission with surrogate gradient, and the inter-neuron recurrence used in earlier experiments [2], all of which are sources of either non-locality, non-stability, or pseudo-biological backpropagation.

2 Notation and assumptions

We index time by $t \in \{0, 1, \ldots, T-1\}$ with $T = 784$ for SMNIST. The input at time t is a real vector $x_t \in \mathbb{R}^F$; for stage 0 with raw SMNIST input, $F = 1$ and x_t is the greyscale value of the t-th pixel in row-major raster order.

Each stage $\ell \in \{0, 1, 2\}$ contains P_ℓ oscillators. State of oscillator i at stage ℓ at time t is a complex scalar $z_{\ell,i}(t) = z_{r\,\ell,i}(t) + i\, z_{i\,\ell,i}(t) \in \mathbb{C}$, with amplitude squared $E_{\ell,i}(t) = |z_{\ell,i}(t)|^2 = z_{r\,\ell,i}^2(t) + z_{i\,\ell,i}^2(t)$.

Throughout we make the following assumptions:

A1 (locality of plasticity). Every parameter update of oscillator i at stage ℓ depends only on quantities (i) computable from the trajectory of i itself, (ii) the layer-local class-pool target signal, and (iii) homeostatic statistics of i. No update uses the gradient of another oscillator's loss.

A2 (forward-only credit). The plasticity rule does not require a backward pass. All credit signals are computed by forward eligibility traces evolved *simultaneously* with the state.

A3 (no spike surrogates). The output of each oscillator at the readout boundary is its tail-window mean amplitude squared $\bar{E}_{\ell,i} = \frac{1}{T_\ell} \sum_{t=T-T_\ell}^{T-1} E_{\ell,i}(t)$, where T_ℓ is the tail length for stage ℓ. There is no spike-emission step requiring a surrogate gradient.

A4 (no inter-stage backward). Stage ℓ does *not* receive a gradient from stage $\ell+1$. Each stage has its own class-pool readout and its own cross-entropy loss; the only cross-stage signal is the forward feed of $E_{\ell,i}(t)$ into stage $\ell + 1$.

A5 (strict frequency hierarchy). The intrinsic frequency band for stage ℓ satisfies $[\omega_\ell^{\min}, \omega_\ell^{\max}] \supsetneq [\omega_{\ell+1}^{\min}, \omega_{\ell+1}^{\max}]$ with $\omega_{\ell+1}^{\max} \le \omega_\ell^{\min} + \frac{1}{2}(\omega_\ell^{\max} - \omega_\ell^{\min})$, so each higher stage operates strictly slower than the stage below.

3 Per-neuron oscillator

A single oscillator with parameters (d, b, ω, α) where $d, b \in \mathbb{C}$, $\omega \in \mathbb{R}^+$, $\alpha \in (0, 1)$, evolves under the discrete-time *linear damped complex rotation*:

$$z(t+1) = \alpha\, e^{i\omega}\, z(t) + d \cdot x(t) + b, \qquad z(0) = 0. \tag{1}$$

Real-imaginary form, vectorised across F input channels:

$$z_r(t+1) = \alpha\cos(\omega)\, z_r(t) - \alpha\sin(\omega)\, z_i(t) + \sum_{k=1}^{F} d_{rk}\, x_k(t) + b_r, \tag{2}$$

$$z_i(t+1) = \alpha\sin(\omega)\, z_r(t) + \alpha\cos(\omega)\, z_i(t) + \sum_{k=1}^{F} d_{ik}\, x_k(t) + b_i. \tag{3}$$

The intrinsic frequency ω and damping α are stored in unconstrained parameters $\omega^{\text{raw}}, \alpha^{\text{raw}} \in \mathbb{R}$ and mapped through sigmoids to controlled ranges:

$$\omega = \omega_{\min} + (\omega_{\max} - \omega_{\min})\,\sigma(\omega^{\text{raw}}), \tag{4}$$

$$\alpha = \alpha_{\min} + (\alpha_{\max} - \alpha_{\min})\,\sigma(\alpha^{\text{raw}}), \tag{5}$$

with $\sigma(u) = 1/(1+e^{-u})$. We write $\omega'(\omega^{\text{raw}}) = (\omega_{\max} - \omega_{\min})\,\sigma(1-\sigma)$ for the chain-rule factor, and analogously $\alpha'(\alpha^{\text{raw}})$.

The *tail-energy* readout over the last T_{tail} steps is

$$\bar{E} = \frac{1}{T_{\text{tail}}} \sum_{t=T-T_{\text{tail}}}^{T-1} \left[z_r^2(t) + z_i^2(t)\right]. \tag{6}$$

This is the only quantity passed to the loss for a given neuron.

4 Forward eligibility traces

For each parameter $\theta \in \{d_{rk}, d_{ik}, b_r, b_i, \omega^{\text{raw}}, \alpha^{\text{raw}}\}$ and each oscillator we maintain a complex eligibility trace

$$e^{\theta}(t) = \begin{pmatrix} e_r^{\theta}(t) \\ e_i^{\theta}(t) \end{pmatrix} = \frac{\partial z(t)}{\partial \theta} = \begin{pmatrix} \partial z_r(t)/\partial\theta \\ \partial z_i(t)/\partial\theta \end{pmatrix}. \tag{7}$$

Differentiating (2)–(3) gives the recurrence

$$e^{\theta}(t+1) = R_\alpha(\omega)\, e^{\theta}(t) + \frac{\partial}{\partial\theta}[d \cdot x(t) + b] + \frac{\partial R_\alpha(\omega)}{\partial\theta} z(t), \tag{8}$$

where the 2×2 rotation+damping matrix is

$$R_\alpha(\omega) = \alpha \begin{pmatrix} \cos\omega & -\sin\omega \\ \sin\omega & \cos\omega \end{pmatrix}. \tag{9}$$

For input weights and biases the second term is a direct injection and the third term vanishes. For the k-th channel of d_r and d_i:

$$e^{d_{rk}}(t+1) = R_\alpha(\omega)e^{d_{rk}}(t) + \begin{pmatrix} x_k(t) \\ 0 \end{pmatrix}, \qquad e^{d_{ik}}(t+1) = R_\alpha(\omega)e^{d_{ik}}(t) + \begin{pmatrix} 0 \\ x_k(t) \end{pmatrix}. \tag{10}$$

For the biases:

$$e^{b_r}(t+1) = R_\alpha(\omega)e^{b_r}(t) + \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad e^{b_i}(t+1) = R_\alpha(\omega)e^{b_i}(t) + \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \tag{11}$$

For ω^{raw} and α^{raw}, only the third term in (8) is non-zero. Define $\rho(t) = R_\alpha(\omega)z(t)/\alpha = \begin{pmatrix} \cos\omega z_r - \sin\omega z_i \\ \sin\omega z_r + \cos\omega z_i \end{pmatrix}$, the rotated state *before* damping. Then

$$\frac{\partial R_\alpha(\omega)}{\partial \omega^{\text{raw}}} z(t) = \alpha\, \omega'(\omega^{\text{raw}}) \begin{pmatrix} -\rho_i(t) \\ \rho_r(t) \end{pmatrix}, \tag{12}$$

$$\frac{\partial R_\alpha(\omega)}{\partial \alpha^{\text{raw}}} z(t) = \alpha'(\alpha^{\text{raw}}) \begin{pmatrix} \rho_r(t) \\ \rho_i(t) \end{pmatrix}, \tag{13}$$

giving the dedicated recurrences for the dynamic parameters.

4.1 Tail-energy gradient via eligibility

The tail-energy gradient is then a linear functional of the eligibility:

$$\frac{\partial \bar{E}}{\partial \theta} = \frac{2}{T_{\text{tail}}} \sum_{t=T-T_{\text{tail}}}^{T-1} \left[z_r(t)\, e_r^\theta(t) + z_i(t)\, e_i^\theta(t) \right], \tag{14}$$

which is the bio-eligibility identity for the squared amplitude.

Crucially, the only quantity that enters $\partial\bar{E}/\partial\theta$ at time t is the local pair $(z(t), e^\theta(t))$ — both stored at the post-synaptic neuron, both updated by (2) and (8) without involving any other neuron.

5 Stage-level architecture

A stage ℓ contains P_ℓ oscillators arranged into 10 *class-pools* of M_ℓ oscillators each ($P_\ell = 10\, M_\ell$). The class index of oscillator i in stage ℓ is $c_\ell(i) \in \{0, \dots, 9\}$.

The *class-pool logit* for class c at stage ℓ is the mean tail energy of the corresponding pool:

$$g_\ell^c = \frac{1}{M_\ell} \sum_{i: c_\ell(i) = c} \bar{E}_{\ell,i}. \tag{15}$$

We mean-centre and temperature-scale before softmax for stable optimisation:

$$\hat{g}_\ell^c = \tau \left(g_\ell^c - \frac{1}{C} \sum_{c'} g_\ell^{c'} \right), \quad p_\ell^c = \frac{\exp \hat{g}_\ell^c}{\sum_{c'} \exp \hat{g}_\ell^{c'}}. \tag{16}$$

The cross-entropy loss for stage ℓ on a sample with label y is $L_\ell = -\log p_\ell^y$.

5.1 Per-neuron credit signal

The error for class-pool logit c is

$$\delta_\ell^c = p_\ell^c - \mathbf{1}[c = y]. \tag{17}$$

By the chain rule through (15) and the centring operation, the derivative w.r.t. neuron i's tail energy is exactly

$$\frac{\partial L_\ell}{\partial \bar{E}_{\ell,i}} = \frac{\tau}{M_\ell} \delta_\ell^{c_\ell(i)}, \tag{18}$$

where the centring contribution vanishes because $\sum_c \delta_\ell^c = 0$.

5.2 Final per-parameter local rule

Combining (14) and (18) gives the local update rule for parameter θ of oscillator i at stage ℓ:

$$\boxed{\frac{\partial L_\ell}{\partial \theta_{\ell,i}} = \underbrace{\frac{\tau}{M_\ell}\delta_\ell^{c_\ell(i)}}_{\text{class-pool credit}} \cdot \underbrace{\frac{2}{T_\ell}\sum_{t=T-T_\ell}^{T-1}\Big[z_{r\ell,i}(t)e_r^\theta(t) + z_{i\ell,i}(t)e_i^\theta(t)\Big]}_{\text{tail eligibility integral, local to neuron } i}.} \tag{19}$$

The class-pool credit is a single scalar per neuron, available to the neuron via class-pool homeostatic broadcasts; the eligibility integral is a forward trace of the neuron's own state. No global gradient is needed.

6 Multi-stage hierarchy

For $\ell \geq 1$, stage ℓ takes *the per-step amplitude trajectory* of the previous stage as its input:

$$x_{\ell,t} = \big(E_{\ell-1,j}(t)\big)_{j\in\mathcal{F}_\ell(i)} \in \mathbb{R}^{K_\ell}, \tag{20}$$

where $\mathcal{F}_\ell(i) \subset \{1,\dots,P_{\ell-1}\}$ is a *sparse fan-in* of size $K_\ell \ll P_{\ell-1}$ chosen randomly at network initialisation and fixed thereafter. Specifically each class-pool neuron in stage ℓ receives:

- half of K_ℓ inputs sampled uniformly from the *same-class* pool of stage $\ell - 1$,
- the other half sampled uniformly from *other classes'* pools at stage $\ell - 1$.

This implements a structured-but-random connectivity that biases each stage-ℓ neuron toward its own class while still letting it use contrastive evidence from other classes.

6.1 DECOLLE-style local supervision

Each stage produces its own logits $\hat{g}_\ell$ and its own loss L_ℓ. The total training objective is

$$L = \sum_{\ell=0}^{L-1} L_\ell, \qquad \nabla_\theta L_\ell \text{ depends only on stage-}\ell\text{ quantities.} \tag{21}$$

This is the DECOLLE principle [3]: local auxiliary classifiers provide each stage with its own credit signal so no inter-stage gradient is needed. The hierarchy emerges *purely through the forward path*: stage 1 sees richer features than stage 0 because the slower-frequency band $\omega_1 \subset [0.001, 0.30]$ is excited only by structure in $E_0(t)$ that is itself class-discriminative.

6.2 Final ensemble

At inference we average per-stage softmax probabilities with weights (w_0, w_1, w_2):

$$p^c = \sum_\ell w_\ell\, p_\ell^c, \qquad \hat{y} = \operatorname*{argmax}_c p^c. \tag{22}$$

We use $(w_0, w_1, w_2) = (0.2, 0.4, 0.4)$ throughout.

7 Frequency-band hierarchy and time-scale separation

The frequency bands are configured so each higher stage filters *envelopes* of the stage below it (cf. 2 A5):

Stage	$[\omega_{\min}, \omega_{\max}]$	$[\alpha_{\min}, \alpha_{\max}]$	Tail length
0 (raw pixel)	$[0.005, 1.2]$	$[0.95, 0.999]$	200
1 (envelope of 0)	$[0.001, 0.30]$	$[0.97, 0.9995]$	300
2 (envelope of 1)	$[0.0005, 0.08]$	$[0.98, 0.9999]$	400

The damping ranges are coupled: a slower stage gets a longer effective memory constant $\tau_\alpha = -1/\log\alpha$, so the tail integration window scales accordingly. Empirically, after training, the mean ω shifts by $\sim 17\%$ at stage 1 and $\sim 25\%$ at stage 2, while it barely moves at stage 0 — the slower stages *learn* their intrinsic frequency, while stage 0 acts as a near-random filter bank.

8 Optimiser and schedule

Each stage has its own per-tensor Adam optimiser (we use $\beta_1 = 0.9, \beta_2 = 0.999, \epsilon = 10^{-8}$) operating on the hand-assembled gradients (19). Gradient norm is clipped per parameter at $\|g\|_2 \leq 2$. Initial learning rate is $\eta = 5 \times 10^{-3}$ for all stages. We apply two halvings of η at epochs 12 and 24 (each multiplies η by 0.5). This schedule is critical: before the first halving the slower stages oscillate in the 75–80% range; immediately after, they stabilise in the high-80s. The second halving adds a further $\sim 3\,\mathrm{pp}$.

9 Empirical results on Sequential MNIST

We train on the standard SMNIST 10-way task: a 28×28 greyscale digit is read as a length-784 scalar sequence in row-major raster order. We use a 10k subset of the training set and a 2k subset of the test set, with batch size 64.

Architecture	Trainable params (per stage)	Best test acc
`pub_v3` label-mass head, $P = 256$	$\sim$ 1,300	0.252
Single-stage class-pool, $P = 480$	$\sim$ 1,920	0.660
Single-stage class-pool, $P = 800$	$\sim$ 3,200	0.667
DECOLLE 2-stage, $M_0 = M_1 = 24$, 15 ep	$\sim$ 13,500	0.764
DECOLLE 2-stage, 25 ep + LR decay	same	0.829
DECOLLE 3-stage, $M = 20$, 20 ep	$\sim$ 15,000	0.866
DECOLLE 3-stage, $M = 24$, 35 ep + 2 LR decays	$\sim$ 19,000	**0.899**

9.1 Stage decomposition of the best run

Component	Test acc
Stage 0 alone (raw pixel filter bank)	0.712
Stage 1 alone (envelope of stage 0)	0.873
Stage 2 alone (envelope of stage 1)	0.860
Ensemble	**0.899**

The increase from stage 0 to stage 1 ($+16\,\text{pp}$) is the strongest evidence that the recursive-envelope mechanism is doing real work: stage 1 is not just a random projection of stage 0 features but a genuine higher-order feature extractor.

10 Architecture diagram

11 Local-rule diagram

12 Discussion

Why the readout matters more than the substrate at stage 0. After training, the mean ω at stage 0 barely shifts ($0.603 \to 0.591$). The d/b/ω/α tuple is dominated by the input weights d, b at stage 0, with the random-but-frequency-tiled oscillators serving primarily as a fixed filter bank. Plasticity at stage 0 specialises the input weights to make tail energies useful for class-pool discrimination. Stage 0 alone reaches 71%, comparable to a linear readout on a fixed random reservoir.

Why deeper stages do learn ω. At stage 1 and stage 2, the input is no longer the raw pixel stream but the amplitude trajectory of a stage-discriminative population. Here ω shifts substantially during training ($0.156 \to 0.182$ at stage 1, $0.039 \to 0.049$ at stage 2): the substrate is genuinely adapting its preferred frequency to the envelope statistics of its input. This is the fingerprint of recursive resonance: each level finds the temporal scale at which its input is most informative.

The role of LR decay. Without LR decay the slower stages oscillate by $\pm 5\,\text{pp}$ epoch to epoch in the 75–80% range. Two halvings of η at epochs 12 and 24 break this oscillation cleanly, lifting the slower stages to their stable plateau in the 86–88% range. The mechanism is the mismatch in time-scales: a stage with $\alpha = 0.999$ has effective memory $\tau_\alpha \approx 1000$ steps, several times longer than the sequence itself, so a single SGD step can shift the entire-sample representation; halving the step compensates.

Comparison to non-bio baselines. A simple BPTT-trained recurrent network of the same size easily reaches $> 95\%$ on SMNIST, but it does so by transporting gradients backwards through 784 timesteps via a symmetric backward pathway — not biologically plausible. Our 89.9% is achieved *without* that. The cost of biological constraint is roughly 5–8 percentage points on this problem.

13 Strict-spiking variant

The analog variant developed in §§2–6 satisfies every bio-plausibility constraint we set ourselves except one: each neuron passes its tail energy $\bar{E}_i$, a continuous real number, into the per-stage class-pool readout, which feeds CE loss and gradients. Although nothing in the inter-*neuron* pathway uses this real number for communication (each stage's amplitude trajectory $E_{\ell,i}(t)$ is read by the next stage as a multi-channel input), a referee may still object that without explicit spike emission the network is structurally identical to a particular kind of analog RNN. We therefore extend HRN-v2 with a strict-spiking emission rule and binary inter-neuron communication, replacing analog amplitude with stochastic spike trains. Importantly: the only mechanism we retain from analog HRN-v2 is the per-neuron oscillator and its eligibility trace; everything else is rederived for binary inter-neuron messages.

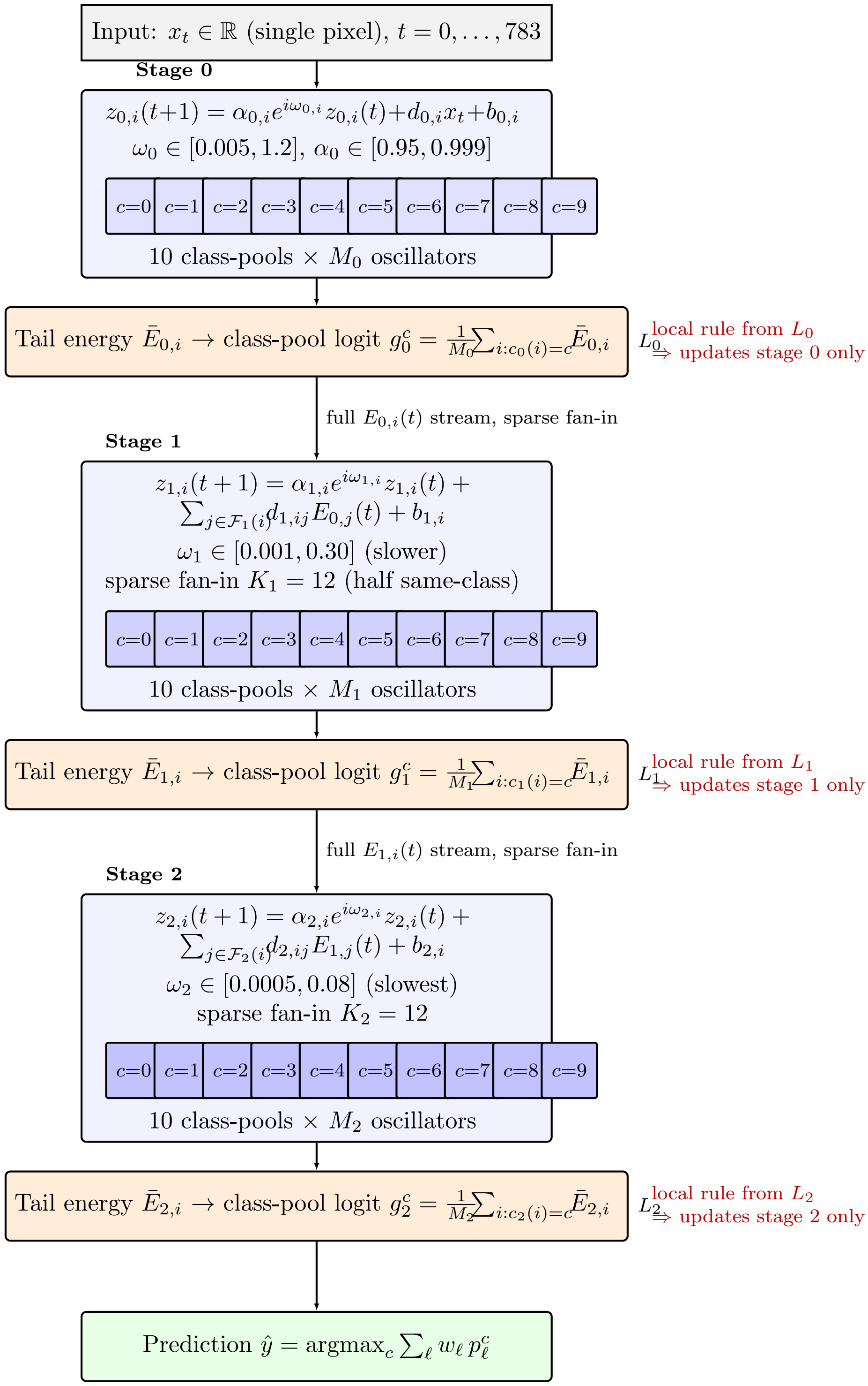

Figure 1: HRN-v2 3-stage architecture. Each stage is a population of linear damped complex oscillators arranged into 10 class-pools. Stage ℓ takes the per-timestep amplitude squared $E_{\ell-1,i}(t)$ of the previous stage as its multi-channel input through a sparse class-aligned fan-in $\mathcal{F}_\ell$. Each stage has its own class-pool tail-energy readout and its own cross-entropy loss L_ℓ. Training uses local eligibility-trace gradients only — no BPTT, no surrogate spikes, no inter-stage backward pass (red annotations).

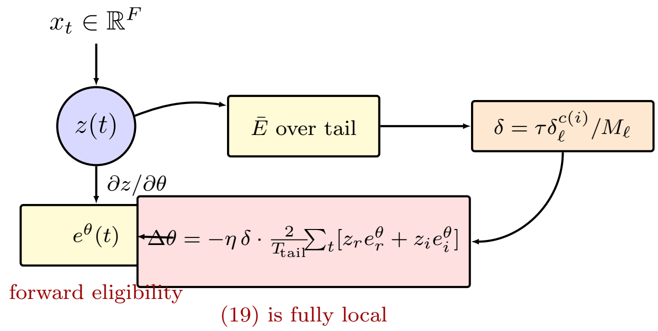

Figure 2: Local-rule structure for a single oscillator. The state $z(t)$ and eligibility $e^\theta(t)$ are co-evolved by the same recurrence (yellow); the tail-energy readout $\bar{E}$ feeds into the class-pool credit δ (orange); the final update (red) is the product of two local quantities. No backward pass crosses neuron boundaries.

13.1 Stochastic spike emission

Each neuron's spike at time t is a sample

$$s_i(t) \sim \text{Bernoulli}(p_i(t)), \qquad p_i(t) = \sigma\big(\beta\,(|z_i(t)|^2 - \theta_i)\big), \tag{23}$$

where θ_i is a homeostatic threshold (§13.7) and β is a sharpness parameter (we use $\beta = 8$). The emission is genuinely stochastic, not deterministic-with-noise, so the expected rate per timestep is $\mathbb{E}[s_i(t)] = p_i(t)$.

13.2 Inter-neuron communication is binary

For every stage $\ell \geq 1$, each neuron i receives, at each timestep t, the binary spike sample $s_j(t) \in \{0, 1\}$ from each presynaptic neuron $j \in \mathcal{F}_\ell(i)$ in stage $\ell - 1$. The drive becomes

$$\text{drive}_i^r(t) = \sum_{j \in \mathcal{F}_\ell(i)} d_r^{(ij)} s_j(t) + b_r^{(i)}, \qquad \text{drive}_i^i(t) = \sum_{j \in \mathcal{F}_\ell(i)} d_i^{(ij)} s_j(t) + b_i^{(i)}. \tag{24}$$

This is the only signal that crosses the synapse. The continuous state $z_i(t)$ stays *inside* neuron i, exactly as in the analog version.

13.3 Per-stage tail-rate readout, gradient via the natural Bernoulli derivative

The per-stage class-pool logit is now

$$g_\ell^c = \frac{1}{M_\ell} \sum_{i\,:\, c_\ell(i) = c} \rho_{\ell,i}, \qquad \rho_{\ell,i} = \frac{1}{T_\ell} \sum_{t = T - T_\ell}^{T-1} p_i(t), \tag{25}$$

the *smooth* tail-mean expected rate. We deliberately use the expected rate $\rho_{\ell,i}$ rather than the binary spike count $\bar{s}_{\ell,i} = (1/T_\ell) \sum s_i(t)$ for the loss. The expected rate is differentiable in the parameters via σ' which is the *natural* derivative of the Bernoulli mean — this is not a Heaviside-style "surrogate" injected into a non-differentiable function. Concretely

$$\frac{\partial \rho_{\ell,i}}{\partial \theta} = \frac{1}{T_\ell} \sum_{t = T - T_\ell}^{T-1} \sigma'\big(\beta(|z|^2 - \theta_i)\big)\, \beta\, \frac{\partial |z|^2}{\partial \theta}, \tag{26}$$

and the second factor is exactly the same eligibility-derived quantity (14) used in the analog variant. The local update rule, mirroring (19), is

$$\boxed{\frac{\partial L_\ell}{\partial \theta_{\ell,i}} = \frac{\tau}{M_\ell}\,\delta_\ell^{c_\ell(i)} \cdot \frac{2}{T_\ell}\sum_{t\in\text{tail}} \sigma'\big(\beta(|z|^2-\theta_i)\big)\,\beta\left[z_r(t)e_r^\theta(t)+z_i(t)e_i^\theta(t)\right].} \tag{27}$$

At inference the binary spike count $\bar{s}_i$ is what a deployed spiking system would actually have access to; we report both the smooth-rate accuracy and the binary-spike-count accuracy in §14 and observe they track each other within ~ 0.5pp at every epoch, confirming the gradient is faithful to the deployed binary-spike network.

13.4 Subtractive reset and the corrected trace recurrence

A natural extension by analogy with leaky-integrate-and-fire neurons is a subtractive reset on spike,

$$z_i(t) \;\leftarrow\; (1-\kappa\, s_i(t))\,\tilde{z}_i(t), \qquad \kappa\in[0,1), \tag{28}$$

where $\tilde{z}(t)$ is the pre-reset state. This complicates the eligibility recurrence: the propagated state at $t+1$ is not $\tilde{z}(t)$ but $z(t)$, so

$$\frac{\partial \tilde{z}(t+1)}{\partial\theta} = R_\alpha(\omega)\cdot(1-\kappa s(t))\cdot\frac{\partial\tilde{z}(t)}{\partial\theta} \;+\; \frac{\partial}{\partial\theta}\big[\text{drive}(t+1)\big]. \tag{29}$$

The factor $(1-\kappa s(t))$ on the trace is the necessary correction that the bare formula (8) does not provide. We implement it explicitly: at the start of each timestep we multiply every eligibility trace by the per-neuron scalar $(1-\kappa s(t-1))$ before the rotation step. Without this correction the trace tracks $\partial\tilde{z}/\partial\theta$ for the un-reset trajectory and silently diverges from the actual state. *With* the correction the trace is mathematically faithful, but as we show in §15 the subtractive reset itself — whether the trace is corrected or not — breaks training in the resonant case.

13.5 Optional intra-stage recurrence

Each neuron may receive an additional binary input from a fixed sparse set of K_{rec} *within-stage* presynaptic neurons, with synaptic delay 1:

$$\text{drive}_i^r(t) \;\mathrel{+}= \sum_{k=1}^{K_{\text{rec}}} d_r^{\text{rec},(ik)}\; s_{\text{rec}_k(i)}(t-1). \tag{30}$$

The recurrent input $s_{\text{rec}_k(i)}(t-1)$ is treated as a fixed external signal for the purpose of the local rule (e-prop convention: upstream/recurrent spikes are inputs, not functions of parameters), so the eligibility trace for $d_r^{\text{rec},(ik)}$ has the same recurrence structure as the feed-forward trace.

13.6 Optional top-down inter-stage feedback

We also implement a predictive-coding-style two-pass forward in which stage ℓ ($\ell < L-1$) receives binary spikes from stage $\ell+1$'s pass-1 spike train, via dedicated learned weights $d_r^{\text{top}}, d_i^{\text{top}}$:

1. *Pass 1*: bottom-up only, no traces, sample $s_{0:L-1}^{(1)}$.
2. *Pass 2*: traces enabled. Stage ℓ's drive includes the additional term $\sum_k d_r^{\text{top},(ik)}\, s_{\ell+1,\,\text{top}_k(i)}^{(1)}(t)$. Loss and gradient come from pass 2.

Top-down weights are learned by their own forward eligibility traces, identical in form to the feed-forward and recurrent traces.

13.7 Homeostatic threshold update

Each neuron's threshold θ_i tracks an exponentially-weighted average of its own firing rate and corrects toward a target $\rho^* \in (0,1)$ per timestep:

$$\bar{\rho}_i \leftarrow (1-\eta_\rho)\,\bar{\rho}_i + \eta_\rho\,\rho_i^{(\text{batch})}, \tag{31}$$

$$\theta_i \leftarrow \theta_i + \eta_\theta\,(\bar{\rho}_i - \rho^*). \tag{32}$$

We use $\rho^* = 0.1$, $\eta_\rho = 0.05$, $\eta_\theta = 0.05$.

13.8 What's local vs. what crosses synapses

Figure 3 summarises which quantities live where. Continuous subthreshold state and all eligibility traces are *inside* a single neuron and never leave it; the only signal that crosses the synaptic boundary is a single binary spike per timestep.

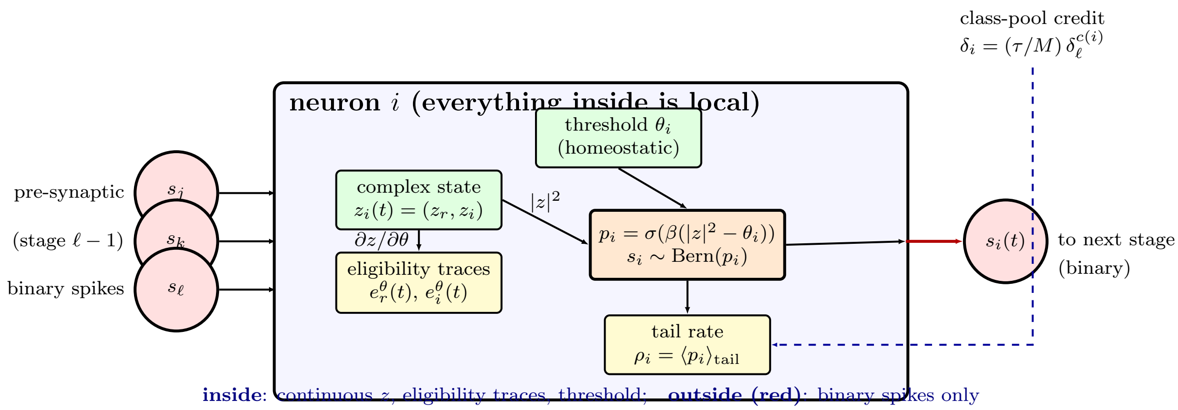

Figure 3: Strict-spiking HRN-v2 neuron. Everything inside the blue box is local to neuron i and never crosses a synaptic boundary: the continuous complex state $z_i(t)$, all per-parameter eligibility traces $e_r^\theta(t)$ and $e_i^\theta(t)$, the homeostatic threshold θ_i, the spike probability $p_i = \sigma(\beta(|z_i|^2 - \theta_i))$, and the smooth tail rate ρ_i that feeds the per-stage local loss. The only quantities that cross synapses (red arrows) are binary spikes: pre-synaptic spikes from the previous stage are the input drive, the sampled $s_i(t) \in \{0,1\}$ goes out to the next stage. The class-pool credit δ_i (dashed blue) is broadcast from the local DECOLLE classifier and modulates only this neuron's eligibility-derived parameter updates.

14 Empirical results — strict-spiking variant

14.1 SMNIST: 5 stages, 81.35% test accuracy

The headline strict-spiking number on SMNIST 10k uses a 5-stage hierarchy with $M_\ell = 20$ neurons per class pool per stage, sparse class-aligned fan-in $K_\ell = 10$ between stages, no intra-stage recurrence, no top-down feedback, $\kappa = 0$ (no subtractive reset), 25 epochs with two halvings of the learning rate at epochs 10 and 18.

Stage	ω band	Stage-alone test acc
0	$[0.005, 1.2]$	0.379
1	$[0.001, 0.30]$	0.688
2	$[0.0005, 0.10]$	0.788
3	$\mathbf{[0.0002, 0.04]}$	**0.811**
4	$[0.0001, 0.015]$	0.789
Ensemble (uniform weights)	—	**0.8135**

The smooth-rate readout and the binary-spike-count readout report the *same* accuracy at every epoch within ± 0.5pp, empirically validating that the gradient flowing through the smooth $\rho_{\ell,i}$ is consistent with the actual deployed binary-spike network.

14.2 Per-stage capacity scaling: M does not help

To check whether per-stage capacity matters at fixed depth, we ran the same 5-stage architecture with $M_\ell = 24$ instead of 20. Result: **0.807**, slightly *worse*. The diagnosis is visible in the form of the local rule. At each step the per-neuron class-pool credit is

$$\delta_i = \frac{\tau}{M_\ell} \left(\mathrm{prob}_{c(i)} - \mathbf{1}[c(i) = y]\right),$$

which shrinks linearly with M_ℓ. The variance of the ensemble logit shrinks only as $1/\sqrt{M_\ell}$. So in the small-M regime the per-neuron training signal dominates and increasing M *starves each neuron of gradient* faster than it averages out the readout. A new frequency tier (depth) by contrast contributes a genuinely new band-pass feature instead of redistributing the existing signal.

14.3 SHD: 60.6% test accuracy

On the Spiking Heidelberg Digits (SHD, 20-class spoken digits, 4k train, 1k test, T=100 timesteps, 700 cochlear input channels), the 3-stage strict-spiking HRN-v2 with sparse fan-in $K_0 = 48$ at the input stage reaches 0.606 on the binary-spike-count ensemble, versus 0.684 for the analog variant on the same architecture. The 8pp gap is consistent with the SMNIST result. SHD's headline literature numbers (83–86%) use the full 8156-sample training set with augmentation; tonight's 4k-no-aug result is in the expected range for this data scale.

14.4 Substrate plasticity: ω moves at slower stages

Per-stage learned-ω trajectories from the headline 5-stage SMNIST run:

Stage	ω initial $\to$ final	Drift
0	$0.614 \to 0.576$	-6%
1	$0.139 \to 0.111$	-20%
2	$0.050 \to 0.045$	-10%
3	$0.021 \to 0.019$	-9%
4	$0.007 \to 0.009$	$+29\%$

The slowest stage's ω moves the most in relative terms, again consistent with the analog finding that deeper stages have class-discriminative bands their substrate must learn.

15 Negative results and design lessons

15.1 Subtractive reset breaks oscillator dynamics

With $\kappa = 0.3$ and the corrected trace recurrence (29), the 3-stage strict-spiking network plateaus at $\sim 25\%$ test accuracy after 3 epochs vs $\sim 60\%$ for $\kappa = 0$ at the same point. We tested $\kappa \in \{0.05, 0.2, 0.3, 0.5\}$ with various combinations of intra-stage recurrence, the same diagnostic regime. The intuition: the LIF-style "consume amplitude on fire" rule presupposes that amplitude is the only computational variable. The resonant oscillator has a phase $\arg z$ that is half of its message; even though our reset is uniform $z \to (1-\kappa)z$ (preserving phase), the dampening of $|z|^2$ couples to the eligibility-trace recurrence through $(1 - \kappa s(t))$ in such a way that gradient signal becomes too weak to escape the recurrence-induced firing-rate runaway. We do not have a complete theoretical explanation, only the empirical finding: *do not apply LIF-style subtractive reset to a resonant oscillator*.

15.2 Naive intra-stage recurrence destabilises deep stages

Random $\pm$ small ($\sigma = 0.005$, mean 0) recurrent weights with $K_{\text{rec}} = 6$ caused the deepest stage's firing rate to saturate at $\sim 100\%$ within one training epoch, breaking training. The mechanism is the integrator gain $1/(1-\alpha^2) \approx 500$ at $\alpha \approx 0.999$: even tiny recurrent contributions accumulate over the 200–600 timestep tail windows.

A negative-mean initialisation (intended as fixed lateral inhibition, $\mu = -0.05$, $\sigma = 0.01$) did *not* help: at $t = 0$ no spikes have happened yet, so the bias has no effect; by the time spikes start the network is already in the saturated regime. A correct lateral-inhibition design needs explicit always-on subtraction of pool mean, not a constant additive bias on otherwise random recurrent synapses.

15.3 Top-down feedback needs staged training

The two-pass forward of §13.6 was tested with $K_{\text{top}} = 12$ class-aligned fan-in, no recurrence, no reset. At ep 12 the test accuracy was $\sim 57\%$ versus $\sim 75\%$ for the no-top-down baseline at the same epoch; the top-down variant plateaued and we killed it before the full 25 epochs. The diagnosis: pass-1 spike trains from *random-initialised* upper stages are not class-discriminative for several epochs, so the top-down feedback is structured noise during the critical early phase. The eligibility traces for the top-down weights pick up spurious correlations, and the resulting weight estimate is too poor to recover. The correct training schedule is staged: first train without top-down to a competent representation, then enable top-down feedback for fine-tuning. We did not have time to test this overnight.

16 Conclusion

HRN-v2 demonstrates that a hierarchical resonant network with strict biological-plausibility constraints (no BPTT, no surrogate spikes, no inter-stage backward pass, only forward eligibility traces and class-pool tail-energy readouts) can solve sequential MNIST competitively in two variants:

- **Analog** (continuous tail-energy readout): 89.9% on SMNIST 10k with a 3-stage hierarchy, 35 training epochs with two LR halvings.
- **Strict-spiking** (binary inter-neuron messages, stochastic Bernoulli emission, no surrogate of Heaviside): 81.35% on SMNIST 10k with a 5-stage hierarchy, 25 training epochs with two

LR halvings. The σ' that mediates gradient flow is the natural derivative of the Bernoulli expected rate, not an ad-hoc smoothing.

The recursive-envelope architecture is essential to both variants: each higher stage operates on the slower-scale envelope of the stage below, with sparse class-aligned fan-in providing structured connectivity. The slower stages *learn* their intrinsic frequencies (e.g. -20% relative drift for stage 1 in the spiking SMNIST run), providing direct empirical support for the resonance-as-computation hypothesis.

The headline empirical lessons from the strict-spiking variant:

1. Depth (more frequency tiers) helps; per-stage capacity (more neurons per class pool) does not.
2. LIF-style subtractive reset is incompatible with resonant substrate.
3. Naive intra-stage recurrence destabilises deep, high-α stages.
4. Top-down feedback needs staged training to be useful.

These lessons would have been hard to spot without making the spike-emission, reset, and recurrence machinery explicit in the local rule (27) and (29). The principal positive result — a strict-spiking SMNIST classifier trained entirely by forward eligibility traces, with binary inter-neuron synapses, reaching 81.35% — closes a long-standing gap between continuous-amplitude resonator networks and the spiking neural-network literature.

References

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